

Metric reconstruction in a Bondi-like gauge

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Metric perturbations obtained from a Hertz potential in an ingoing radiation gauge generically grow linearly in an outgoing coordinate r at future null infinity, and are therefore not asymptotically flat in the Bondi sense. Hollands and Toomani showed that this growth is pure gauge and can be removed, within the Green–Hollands–Zimmerman (GHZ) framework, by a residual gauge vector together with a carefully chosen choice of the undetermined integration “constants” of the Hertz potential. In this work we turn that formal construction into a practical, mode-based algorithm. We show that this entire completion is generated by a single “Held Master Scalar” (HMS), which can be thought of as a function on future null infinity. The HMS is obtained from the leading asymptotic piece of the ψ_0 Weyl scalar through a Teukolsky–Starobinsky inversion. We give explicit (ℓ, m, ω) formulas for the four integration constants of the Hertz potential associated with ψ_0 , and for the residual gauge vector, thereby bringing the metric perturbation in a Bondi-like gauge. In Schwarzschild the entire construction is diagonal in spin-weighted spherical harmonics and every completion field is available in closed form; in Kerr we give the corresponding (ℓ, m, n, k) scheme, in which the only additional ingredients are the spheroidal-to-spherical mixing coefficients and a tridiagonal “ $\cos \theta$ coupling matrix”. We have numerically implemented the full scheme, including the corrector piece of the GHZ framework, and demonstrate the cancellation of the growing radiation-gauge terms and the recovery of Bondi falloff. We discuss how our approach fits into the puncture scheme for self-force calculations.

I. INTRODUCTION

Large-mass-ratio inspirals of a comparatively small companion into a black hole present interesting conceptual mathematical challenges, such as the problem of scaling solutions, described from a physical perspective first in [REF], and more recently [REF] in a limited scenario permitting complete mathematical control. Because such systems are among the principal targets of the Laser Interferometer Space Antenna there is also a very strong motivation for developing accurate models capable of describing the dynamical evolution of the binary, and the emitted gravitational waves, over a very large number of orbits. The large mass ratios of up to $10^5 - 10^6$ expected in practice are so far inaccessible to fully numerical schemes [REF], in particular since a quite large number of such simulations corresponding to varying orbital parameters would be needed for wave-form modelling [REF].

The current state of the art are instead perturbative, semi-analytical methods within the self-force scheme [REFs]. In this scheme, the near field of the small object is typically isolated at various orders in perturbation theory—in practice second order is considered necessary. The residual field, i.e., the gravitation field with the small-object field extracted, represents the large-scale “trend” of the gravitational field as opposed to the—often infinite—local perturbation, and is thought of as responsible for the self-force on the particle [REFs]. This self-

force gives rise to a gradual change of the orbital parameters, which must be updated throughout the dynamical evolution. Since the gradual change is very slow compared to the time-scale of the gravitational waves ($\gtrsim$ orbital period), it is possible to separate the problem into an evolution equation for the orbital parameters governed by a slow time, and perturbative calculations, at various orders, in a *fixed* Kerr background geometry and *fixed* orbital parameters, governed by a fast time[? ?] [further REFS].

In this paper, we focus on this second problem, which is the main input into the first. In each perturbative order, our problem, therefore, is basically to find a solution, h_{ab} , to the linearized Einstein equation $\mathcal{E}h_{ab} = T_{ab}$ on a fixed Kerr background, where $\mathcal{E}$ is the linearized Einstein operator¹. T_{ab} typically represents the stress tensor at the respective order. In case h_{ab} represents the residual field at that order, T_{ab} will include, besides the point-particle stress tensor associated with the smaller object itself, also pieces arising from the lower order residual metric perturbations, and with the subtraction pieces of the near field (“punctures” [REF]), up to the same order. The latter are determined entirely locally [REF] and, while complicated, do not pose an essential problem.

There are various approaches to solving $\mathcal{E}h_{ab} = T_{ab}$ in practice, see [REFs]. A common benchmark criterion is that one would like an h_{ab} in a well-behaved gauge². Here, “well behaved” means e.g., that one would not like

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¹ In our conventions, it is given by [PZ: [Eq. ?] of ?] — the \cite here had an empty key and was aborting the build.

² The linearized Einstein operator is gauge invariant, so a gauge piece represents the part of the non-linear gauge transformation at the perturbation order that one is working in.

the metric to contain artificial singularities, or a non-decaying behavior at null-infinity, non-smooth behavior at the horizon, etc. In particular, it is highly preferable to find h_{ab} in a Bondi-like gauge, such that asymptotic quantities like BMS-charges, memory, or fluxes are readily identifiable, or such that the contribution of h_{ab} to the stress tensor at the next—in practice second—perturbation order is manageable.

In the present paper, we focus on the metric reconstruction method, originally due to [REF] in the vacuum ($T_{ab} = 0$) case, see [REF] for potentially viable alternatives. Even though, as our application, we will consider a concrete puncture scheme yielding the residual part of the metric perturbation from an “effective” stress energy tensor T_{ab} at first order, and for a circular equatorial orbit of Kerr, our general scheme is conceptually independent of such restrictions as long as T_{ab} has sufficient, and standard for radiating sources, decay (6) at future null infinity.

The broad aim of reconstruction methods is to obtain the metric perturbation h_{ab} from the spin $s = +2$ Weyl scalar ψ , which obeys Teukolsky’s decoupled separable equation for each mode [?]. Such mode decompositions are particularly suitable for systems with periodic sources representing orbiting point-like companions. From ψ , one then obtains a so-called Hertz potential, Φ , by a kind of inversion [REF], and then the metric perturbation as $h_{ab} = 2 \operatorname{Re} \mathcal{S}_{ab}^\dagger \Phi$, where $\mathcal{S}_{ab}^\dagger$ is a tensor-valued “reconstruction operator” [REF]. Actually, this scheme by itself is inconsistent for generic non-trivial sources T_{ab} , but it can be adapted by adding a certain correction piece, x_{ab} , to $2 \operatorname{Re} \mathcal{S}_{ab}^\dagger \Phi$ [REF]. The construction of x_{ab} is called the GHZ [REF] method. A practical way to combine the GHZ scheme with the puncture method [REF], i.e., for determining the residual metric perturbation, was proposed later in [REF], and implemented for a circular orbit in Schwarzschild in [REF]. The approach [REF] nicely avoids a stringlike gauge singularities emanating from the source, but it does not by itself produce a metric perturbation in a Bondi-like gauge.

While far from obvious, it is physically plausible that h_{ab} actually *has* an asymptotically flat gauge representative. However, even if it does (which we assume is the case and provide evidence for³) it is non-trivial to actually find this gauge representative, see [REF] for one possible scheme. In this paper, we instead follow a recent extension of the GHZ method due to [REF], which delivers a gauge vector field ζ^a (and at the same time a particular choice of the non-unique Hertz-potential Φ) achieving a *Bondi-like gauge*. The total metric perturba-

tion is hence

$$h_{ab} = 2 \operatorname{Re} \mathcal{S}_{ab}^\dagger \Phi + x_{ab} - \mathcal{L}_\zeta g_{ab} + \dot{g}_{ab} \quad (1)$$

wherein g_{ab} is the Kerr metric, and $\dot{g}_{ab}$ a metric perturbation corresponding to an infinitesimal change⁴ ($\dot{M}, \dot{a}$) of the Kerr parameters (M, a). Because the reconstructed piece $2 \operatorname{Re} \mathcal{S}_{ab}^\dagger \Phi$ is in ingoing radiation gauge (IRG), and because ζ^a turns out to be a “residual” i.e., IRG compatible gauge vector field [REF], the entire expression (1) is in an IRG. Our Bondi-like gauge means this, plus the remaining four decay conditions (7) on the null-tetrad components of h_{ab} .

Building on [REF], we will present a remarkably simple algorithm for obtaining *both* ζ^a , and the particular choice of Φ . This algorithm is as follows:

- (i) Find the leading order asymptotic piece ψ^{5° of the “0” perturbed-Weyl-scalar component ψ (the r^{-5} piece), which is used to
- (ii) Find a certain “Held Master Scalar” (HMS) f° , where a $\circ$ indicates a quantity that is intrinsic to future null infinity, or more pragmatically, a function of u, θ, φ in an outgoing Eddington-like coordinate system. f° is determined from ψ^{5° by an angular Teukolsky-Starobinsky relation.
- (iii) Apply certain explicit geometrical differential operators to our HMS f° , to obtain both the leading pieces $\Phi^{3^\circ}, \dots, \Phi^{0^\circ}$ (the $r^3, \dots, r^0$ -pieces of Φ , which are undetermined from the inversion relation linking ψ_0 and Φ), and the components of the residual gauge vector field ζ^a .

Since the corrector x_{ab} is found by three simple, transport equations [REF] with explicit Green’s function [REF] if we want, and since it is known [REF] how to solve the inversion problem $\psi \rightarrow \Phi$, the problem of finding h_{ab} as a solution to $\mathcal{E}h_{ab} = T_{ab}$ in reconstructed form, ingoing radiation gauge, *and* Bondi-like form near future null infinity is solved.

In this work, we also use the separability of ψ_0 (and consequently, of Φ), to devise a concrete mode scheme for (i)–(iii). Actually, we can consider both a so-called “ m -mode” scheme (factoring out only the $e^{-i\omega t + im\varphi}$ -dependence of the solutions), or a full “ (ℓ, m) -mode” scheme (factoring out also the spheroidal harmonics). In general, our implementation architecture for determining the corrector x_{ab} supports both schemes. However, since an m -mode scheme is currently unavailable for solving the spin ± 2 Teukolsky equation, we carry out the remaining steps in an (ℓ, m) -mode scheme. Nevertheless, we think that all of our methods could be used in an m -mode scheme once ψ_0 would be given as a numerical solution on an (r, θ) -grid covering the exterior of Kerr.

³ For solutions of compact support (in some gauge), this follows from non-trivial mathematical theorems about the falloff of solutions to the linearized Einstein equation. However, these theorems do not directly apply to the retarded-type solutions under consideration here, wherein a source extending all the way to past time-like infinity is present.

⁴ ($\dot{M}, \dot{a}$) may be found from T_{ab} [REF]. $\dot{g}_{ab}$ may be presented in IRG [REF].

Expectedly, our scheme is particularly straightforward in Schwarzschild-spacetime, but we also describe the necessary changes in Kerr. These have mostly to do with the fact that the angular GHP-like operators do not act as simple raising- or lowering operators unlike in Schwarzschild [REF]. As a consistency check of our scheme, we numerically check that h_{ab} is in a Bondi-like gauge.

In this work, we focus concretely on the problem of finding the first order residual metric perturbation h_{ab}^R within a well-established puncture scheme [REFs], for circular equatorial orbits $\gamma(\tau)$ in Kerr spacetime, so that the concrete problem to be solved is $\mathcal{E}h_{ab}^R = T_{ab}^E$. Here, T_{ab}^E is an effective source taking into account the original point particle source,

$$T^{ab} = m \int d\tau \dot{\gamma}^a(\tau) \dot{\gamma}^b(\tau) \delta^4[\cdot - \gamma(\tau)] / \sqrt{-g}, \quad (2)$$

and additionally a stress tensor corresponding to the singular field near the body, called a “puncture”, h_{ab}^P , i.e.,

$$T_{ab}^E := T_{ab} - \mathcal{E}h_{ab}^P. \quad (3)$$

In concretely constructing T_{ab}^E in our setting, we rely on previous non-trivial works [REF], which we partly generalize to the Kerr case. The full metric perturbation at first order would then be $h_{ab} = h_{ab}^P + h_{ab}^R$, where h_{ab}^R would e.g., be used to compute the self-force near the particle. While we could do this, we will instead mostly concentrate on the asymptotic region where the technical advance of our new method plays out. The full-scheme is in fact basically a no-string approach avoiding a string-type singularity both outside and inside the particle orbit.

We finally note that, having solved the non-trivial problem of constructing T_{ab}^E at higher orders in perturbation theory, our method would directly generalize to this case: We would find ψ_0 , and x_{ab} at that order, and then go through the same steps (i)–(iii).

This paper is organized as follows ...

II. CONVENTIONS FOR KERR BACKGROUND AND DIFFERENTIAL OPERATORS

Our conventions regarding the Kerr-metric, the Newman-Penrose (NP)– and Geroch-Penrose-Held (GHP) formalisms [REFs], are the same as in GHZ [?], [?]. In particular, the signature of the metric is $(+, -, -, -)$. As these references we will actually mainly use Held’s modified GHP calculus [? ?], in the same conventions as stated in [?]. The reader is not expected to have any familiarity with these formalisms since for our concrete application, it is only necessary to recall the definitions of certain coefficient functions and differential operators.

These are stated in the Kinnersley frame, see e.g., [?], [?], and Boyer-Lindquist coordinates called (t, r, θ, φ) ,

or in outgoing Eddington coordinates called $(u = t - r_*, r, \theta, \varphi_*)$ for Kerr. They are related by

$$dr_* = \frac{r^2 + a^2}{\Delta} dr, \quad d\varphi_* = d\varphi - \frac{a}{\Delta} dr, \quad \Delta = r^2 - 2Mr + a^2. \quad (4)$$

In these conventions, the coefficients that we require, are

$$\rho = -\frac{1}{r - ia \cos \theta}, \quad (5)$$

and $\Psi^\circ, \rho'^\circ, \tau^\circ, \Omega^\circ$, and a certain geometric basis of directional derivative operators, see Table I. As a matter of notation, in the Held-formalism, a $\circ$ intrinsically means a GHP scalar that is annihilated by the GHP-operator $\mathbf{P}$; pragmatically this just means that it is independent of r in Eddington coordinates in our conventions.

The quantity s is the spin weight of a GHP scalar or operator. It must be taken into account for every quantity these operators are acting on, and a GHP operator with the indicated weight will shift these by the amount associated with the operator. A nice feature of the above Held operators– and scalars is that any GHP invariant formulas, as given below, is that they are gauge invariant, in the sense that they could readily be converted from the Kinnersley frame to any other one related by a local boost/rotation of this NP tetrad see e.g., [?]. This would be practically useful if we wanted to convert our abstract formulas e.g., to the Hartle-Hawking– or Carter frames, and is also geometrically more satisfying.

III. THE ASYMPTOTICALLY FLAT RECONSTRUCTION SCHEME

Here we describe our asymptotically flat reconstruction scheme extracted from [?, Thm. 4] and its derivation. Theorem [?, Thm. 4] applies rigorously to (smooth) solutions to the linearized Einstein equations (EE) $\mathcal{E}h_{ab} = T_{ab}$, such that

1. The NP-components of the stress energy tensor T_{ab} in our Kinnersley frame behave in Eddington coordinates as:

$$\begin{aligned} T_{nn} &= O\left(\frac{1}{r^2}\right), \\ T_{ln}, T_{m\bar{m}}, T_{nm} &= O\left(\frac{1}{r^3}\right), \\ T_{lm}, T_{mm} &= O\left(\frac{1}{r^4}\right), \\ T_{ll} &= O\left(\frac{1}{r^5}\right), \end{aligned} \quad (6)$$

towards $\mathcal{I}^+$, i.e., as $r \rightarrow \infty$, at fixed (u, φ_*, θ) .

2. There exists *some* gauge such that the NP components of h_{ab} have asymptotic expansions with decay

TABLE I: Summary of Held’s GHP quantities in our conventions, taken from [?]. On a mode with dependence $e^{-i\omega t + im\varphi} = e^{-i\omega u + im\varphi_*}$, one puts $\partial_u \rightarrow -i\omega$ and $\partial_{\varphi_*} \rightarrow im$. $\tilde{\mathbf{D}}$ and $\tilde{\mathbf{D}}'$ are thereby equal to the spin-raising and spin-lowering operators when acting on spin-weighted spherical harmonics; see, e.g., [?].

Held quantity	Value	Spin Weight
τ°	$-\frac{ia \sin \theta}{\sqrt{2}}$	
ρ'°	$-\frac{1}{2}$	
Ψ°	M	
Ω°	$-2ia \cos \theta$	
$\mathbb{P}$	∂_r	
$\tilde{\mathbb{P}}'$	$\partial_u - \frac{1}{2} \frac{r^2 - 2Mr + a^2}{r^2 + a^2 \cos^2 \theta} \partial_r$	
$\tilde{\mathbf{D}}$	$-\frac{1}{\sqrt{2}}(\partial_\theta + i \csc \theta \partial_{\varphi_*} - ia \sin \theta \partial_u - s \cot \theta)$	
$\tilde{\mathbf{D}}'$	$-\frac{1}{\sqrt{2}}(\partial_\theta - i \csc \theta \partial_{\varphi_*} + ia \sin \theta \partial_u + s \cot \theta)$	

orders

$$\begin{aligned} h_{ll}, h_{lm}, h_{ln}, h_{m\bar{m}} &= O\left(\frac{1}{r^2}\right), \\ h_{nn}, h_{nm}, h_{mm} &= O\left(\frac{1}{r}\right), \end{aligned} \quad (7)$$

as $r \rightarrow \infty$, at fixed (u, φ_*, θ) . These condition would be satisfied if one assumes that there exists a Bondi-gauge for h_{ab} , see [? , App. D]. In an IRG, all NP components involving at least on “ l ” would vanish.

3. There exist $\delta M, \delta a$ such that, subtracting a corresponding zero mode (see [? , App. C] for its expression in IRG)

$$h_{ab} \rightarrow h_{ab} - \dot{g}_{ab}, \quad (8)$$

the leading expansion coefficients in the asymptotic expansion (??_exp[PZ: missing eq]) all go to zero as $u \rightarrow -\infty$.

4. At $\mathcal{I}^+$ the Bondi news tensor satisfies the usual Price-law tail [Ref.] as $u \rightarrow \infty$.

Of course in practice one does not know from the outset whether the conditions 2)–4) on the metric perturbation hold unless one can appeal to abstract mathematical theorems to this effect. We are unfortunately not aware of such abstract theorems in the specific self-force setting below, mainly due to the fact that the “particle exists forever into the past”. We therefore need to *assume* that the conclusions [? , Thm. 4] hold, and we extract the following prescription from the proof of [? , Thm. 4], for finding an $x_{ab}, \Phi, \zeta^a, \dot{g}_{ab}$ in the decomposition (1) for h_{ab} . Our perturbation will be in a *Bondi-like gauge*, meaning

by definition IRG, $h_{lm} = h_{ln} = h_{ll} = 0$, and the remaining four conditions in (7). This gauge could easily be converted to an actual Bondi gauge [? , App. D], but for the purposes of extracting quantities associated with gravitational radiation, or for using h_{ab} to build the source at the next perturbative order, the Bondi-like gauge appears just as good.

First, the perturbed Teukolsky scalar

$$\psi \equiv \delta C_{lm} \quad (9)$$

of spin-weight $s = +2$, is determined, through⁵ Teukolsky’s equation [REF],

$$\mathcal{O}\psi = \mathcal{T}^{ab}T_{ab}, \quad (10)$$

where $\mathcal{T}$ is Teukolsky’s source operator, and $\mathcal{O}$ is Teukolsky’s wave operator, see e.g., [? , App. G] for their concrete expressions in our conventions. We assume *retarded* boundary conditions. Then by the peeling theorem⁶

$$\psi = \frac{\psi^{5\circ}}{r^5} + O\left(\frac{1}{r^6}\right) \quad (11)$$

Second, the Hertz potential Φ is obtained from the well-known [REF] inversion relation⁷

$$\partial_r^4 \Phi = 2\bar{\psi}. \quad (12)$$

⁵ This ψ is $-\psi_0$ of [?]; we have no need for ψ_4 in this article, and consequently we drop the subscript “0”.

⁶ This would follow in a rigorous setting by (7); here it is strictly speaking an assumption which we will find to be correct in our application to self-force.

⁷ This equation refers to Eddington coordinates $(u, r, \varphi_*, \theta)$ and the Kinnersley frame. The GHP invariant form of this equation is $\tilde{\mathbb{P}}^4 \Phi = 2\bar{\psi}$.

We then have a unique solution *minimal solution*, i.e., such that $\Phi = O(r^{-1})$. This minimal solution is however *not* the one appearing in the asymptotically flat IRG metric (1). Instead, we have to add to the minimal solution specific terms:

$$\Phi \longrightarrow \Phi + \Phi^{0\circ} + \frac{\Phi^{1\circ}}{\rho} + \frac{\Phi^{2\circ}}{\rho^2} + \frac{\Phi^{3\circ}}{\rho^3}. \quad (13)$$

These terms are *growing* as r^3 for $r \rightarrow \infty$, and this is actually essential: their contribution under the reconstruction operator $\mathcal{S}_{ab}^\dagger$ combines with $\mathcal{L}_\zeta g_{ab}$, in (1), leading furthermore to a delicate cancellation of various subleading terms, and leaving in the end a perturbation with Bondi-like falloff at future null infinity [?, Thm. 4]. For this, a specific choice of the growing terms must of course be made; in fact, by working through the proof of [?, App. J], the entire completion is generated by a single *Held Master Scalar* (HMS) f° of spin weight +2, defined by

$$2\tilde{\mathbf{P}}'^3 \bar{\psi}^{5\circ} = \tilde{\delta}'^4 \tilde{\delta}^4 \bar{f}^\circ - 9\Psi^\circ \bar{\Psi}^\circ \tilde{\mathbf{P}}'^2 \bar{f}^\circ. \quad (14)$$

The four Held integration constants appearing in (13) are determined from the HMS f° by the triangular system

$$\Phi^{3\circ} = -2\tilde{\mathbf{P}}' \bar{f}^\circ, \quad (15a)$$

$$\Phi^{2\circ} = -2\tilde{\delta}' \tilde{\delta} \bar{f}^\circ, \quad (15b)$$

$$\tilde{\mathbf{P}}' \Phi^{1\circ} = -\tilde{\delta}'^2 \tilde{\delta}^2 \bar{f}^\circ - (4\bar{\tau}^\circ \tilde{\delta} + 3\Psi^\circ) \tilde{\mathbf{P}}' \bar{f}^\circ, \quad (15c)$$

$$\begin{aligned} \tilde{\mathbf{P}}'^2 \Phi^{0\circ} = & -\frac{1}{3} \tilde{\delta}'^3 \tilde{\delta}^3 \bar{f}^\circ - \left(2\bar{\tau}^\circ \tilde{\delta}' \tilde{\delta}^2 + \Psi^\circ \tilde{\delta}' \tilde{\delta} \right. \\ & \left. + 2\Psi^\circ \rho'^\circ \right) \tilde{\mathbf{P}}' \bar{f}^\circ. \end{aligned} \quad (15d)$$

Thirdly, we build $\zeta_l^\circ, \zeta_n^\circ, \zeta_m^\circ$ from

$$\bar{\zeta}_m^\circ = \tilde{\delta} \bar{f}^\circ, \quad (16a)$$

$$\tilde{\mathbf{P}}' \zeta_l^\circ = -\text{Re} \left(\tilde{\delta} \tilde{\delta} \bar{f}^\circ \right), \quad (16b)$$

$$\begin{aligned} \tilde{\mathbf{P}}' \zeta_n^\circ = & -\text{Re} \left(\tilde{\delta}' \tilde{\delta}^3 \bar{f}^\circ - \rho'^\circ \tilde{\delta} \tilde{\delta} \bar{f}^\circ - 2\bar{\tau}^\circ \tilde{\delta} \tilde{\mathbf{P}}' \bar{f}^\circ \right. \\ & \left. - \frac{1}{2} \Omega^\circ \tilde{\delta} \tilde{\delta} \tilde{\mathbf{P}}' \bar{f}^\circ \right). \end{aligned} \quad (16c)$$

from the HMS $\bar{f}^\circ$, which give the residual gauge vector field ζ^a [REF], see e.g., [?, App. E]. In fact, for (1), we only need to evaluate $\mathcal{L}_\zeta g_{ab}$ which by the formulas in Appendix A is determined explicitly by $\zeta_l^\circ, \zeta_n^\circ, \zeta_m^\circ$.

Finally, we determine x_{ab} from the GHZ transport equations by a backward integration in r from $\mathcal{I}^+$, see e.g., [?, Sec. 3.1.3], and we determine the zero mode $\dot{g}_{ab}$ by Abbot-Deser type integrals from T_{ab} [REF]. These last steps are actually even easier in our model problem as we will see, since the stress tensor is non-zero only locally near the particle.

We thereby have all pieces in (1) giving us the metric perturbation in a Bondi-like gauge in IRG. For completeness, the spin weights of every object appearing in the construction are collected in Table II.⁸

IV. CIRCULAR ORBIT IN SCHWARZSCHILD

Here we outline our scheme for a point particle on a circular orbit in Schwarzschild. Due to the high symmetry, it is convenient to expand any spin- s quantity ${}_s X$ in this spacetime in modes as

$${}_s X = \sum_{\omega, \ell, m} {}_s X_{\omega \ell m}(r) {}_s Y_{\ell m}(\theta) e^{-i\omega t + im\varphi}, \quad (17)$$

where our conventions for the spin-weighted spherical harmonics ${}_s Y_{\ell m}(\theta)$ [REF] are taken to be the same as [?]. For generic orbits, the possible frequencies are the harmonics of the basic orbital frequencies $(\Omega_r, \Omega_\theta, \Omega_\varphi)$ labeled by three integers. For a circular orbit as considered here, this reduces to $\omega = n\Omega_\varphi, n \in \mathbb{Z}$ with $\Omega_\varphi = \sqrt{M/r_0^3}$, where r_0 is the radius of the orbit.

Following [REF], a puncture h_{ab}^P for such an orbit has been obtained by the method of Ref. [?], where we take the order of the puncture to 3. Then the NP components of the effective stress energy tensor T_{ab}^E (3) are formed using this puncture and the stress energy tensor of the particle (2). As described in [REF], the puncture is first obtained in a certain body-centered coordinate system and is cut off by a window function around r_0 . Then its NP components are transformed to Boyer-Lindquist coordinates as a numerical array corresponding to a discrete grid of (r, θ) -values, as are the NP components of T_{ab}^E . Using semi-analytic tools developed in [REF], one then obtains the (ℓ, m) -mode expansion (17) of the corresponding effective Teukolsky source in (10). Its expansion coefficients are numerically given as an array on a discrete set of r -values per mode [REF]. The order of the puncture governs the fall-off of the mode amplitudes in ℓ .

With these at hand, one obtains the mode expansion coefficients of the perturbed Weyl-scalar ψ (9) by solving the mode version of Teukolsky's equation (10), i.e., the radial Teukolsky equation for the modes coefficients (17) of ψ , with the effective Teukolsky source, and with purely outgoing (at the horizon $r = r_+$) and purely downgoing (at future null infinity $r = \infty$) boundary conditions. We have checked that this ψ satisfies peeling (11), and thereby we can extract the mode-sum coefficients of the leading coefficient,

$$\psi^{5\circ} = \sum_{\omega, m, \ell} \psi_{\omega \ell m}^{5\circ} {}_{+2} Y_{\ell m}(\theta) e^{-i\omega u + im\varphi}, \quad (18)$$

⁸ They fix both the angular matrix elements of Sec. ?? and, through (??), the sign attached to each complex conjugation; the odd-spin entries are the ones for which the factor $(-1)^s$ is not unity.

TABLE II: Spin weights of the GHP and reconstruction fields entering the adjusted reconstruction. The second row is the sign $(-1)^s$ appearing in the conjugation rule $(\bar{X})_{\omega,\ell,m} = (-1)^{s+m} X_{-\omega,\ell,-m}$ for mode sum coefficients (17).

	f°	$\bar{f}^\circ$	Φ^{n°	$\zeta_l^\circ, \zeta_n^\circ$	ζ_m°	$\bar{\zeta}_m^\circ$
Spin weight s	+2	-2	-2	0	+1	-1
$(-1)^s$	+	+	+	+	-	-

TABLE III: Spin weights of the metric perturbation.

	h_{ll}	h_{ln}	h_{nn}	$h_{m\bar{m}}$	h_{nm}	$h_{n\bar{m}}$	h_{mm}
Spin weight s	0	0	0	0	+1	-1	+2
$(-1)^s$	+	+	+	+	-	-	+

by fitting against r^{-5} . Recall that $u = t - r_*$ is the Eddington coordinate, and that quantities with a $\circ$ by construction have no r -dependence. Either by Ori's inversion scheme [REF], or by simply integrating $\bar{\psi}$ four times against r beginning from $r = \infty$ (in practice an r -value much larger than both M and the particle's orbital radius r_0), we get the minimal solution $\Phi = O(r^{-1})$ of the inversion relation (12) for the Hertz-potential, mode-by-mode.

Next, we solve the mode version of our HMS (14). In doing so, we use that $\tilde{\mathbf{P}}' = -i\omega$ in modes, and that the combination of angular operators in (14) is precisely the combination appearing in the Teukolsky-Starobinsky identities—where both observations generalize to Kerr. The mode coefficients $f_{\omega\ell m}^\circ$ of the HMS are thereby found to be:

$$f_{\omega\ell m}^\circ = \frac{2i\omega^3}{D_{\ell m\omega}} \psi_{\omega\ell m}^{5^\circ} \quad (19)$$

where $D_{\ell m\omega}$ is a Teukolsky-Starobinsky constant. Its value in Schwarzschild is $D_{\ell m\omega} = (\ell + 1)\ell(\ell - 1)(\ell - 2)/16 + 9M^2\omega^2$.

Using the ladder operator properties of Held's angular derivatives in terms of spin-weighted harmonics [REF], the four Held completion pieces (15) in the Hertz potential are similarly found to be

$$\Phi_{\omega\ell m}^{0^\circ} = \frac{-\frac{2}{3}i\omega(-1)^m}{D_{\ell m\omega}} \quad (20a)$$

$$\times \left[{}_2B_{\ell 1}B_\ell - 3iM\omega \left(1 + \sqrt{{}_2B_\ell} \right) \right] \bar{\psi}_{-\omega,\ell,-m}^{5^\circ}, \quad (20b)$$

$$\Phi_{\omega\ell m}^{1^\circ} = \frac{2\omega^2(-1)^m}{D_{\ell m\omega}} \left({}_2B_\ell - 3iM\omega \right) \bar{\psi}_{-\omega,\ell,-m}^{5^\circ}, \quad (20c)$$

$$\Phi_{\omega\ell m}^{2^\circ} = \frac{4i\omega^3 {}_2B_{\ell 1}B_\ell (-1)^m}{D_{\ell m\omega}} \bar{\psi}_{-\omega,\ell,-m}^{5^\circ}, \quad (20d)$$

$$\Phi_{\omega\ell m}^{3^\circ} = \frac{-4\omega^4(-1)^m}{D_{\ell m\omega}} \bar{\psi}_{-\omega,\ell,-m}^{5^\circ}, \quad (20e)$$

The residual gauge vector ζ^a 's building blocks (16) needed to compute $\mathcal{L}_\zeta g_{ab}$ by the formulas in App. A

are

$$\zeta_{m,\omega\ell m}^\circ = \frac{-2i\omega^3 \sqrt{{}_2B_{\ell 1}B_\ell}}{D_{\ell m\omega}} \psi_{\omega\ell m}^{5^\circ}, \quad (21a)$$

$$\zeta_{l,\omega\ell m}^\circ = \frac{\omega^2 \sqrt{{}_2B_\ell}}{D_{\ell m\omega}} \psi_{\omega\ell m}^{5^\circ} + \text{c.c.}, \quad (21b)$$

$$\zeta_{n,\omega\ell m}^\circ = \frac{-\omega^2 \sqrt{{}_2B_\ell}}{D_{\ell m\omega}} \left({}_1B_\ell + \frac{1}{2} \right) \psi_{\omega\ell m}^{5^\circ} + \text{c.c.}, \quad (21c)$$

where “+ c.c.” denotes the mode-space real part, i.e. the addition of $(-1)^m \bar{\psi}_{-\omega,\ell,-m}^{5^\circ}$ inside the same prefactor. We also defined $\sqrt{{}_2B_\ell} = \frac{1}{2}\sqrt{(\ell+1)\ell(\ell-1)(\ell-2)}$ and ${}_2B_{\ell 1}B_\ell = \frac{1}{2}(\ell+2)(\ell-1)$.

In our inward integration scheme, the corrector x_{ab} , obtained via the GHZ transport equations, vanishes outside the window $(r_{\min}, r_{\max})$ around the particle (puncture). Thus, outside this window, the reconstructed metric (1) is in this case simply

$$h_{ab} = 2 \text{Re } \mathcal{S}_{ab}^\dagger \Phi - \mathcal{L}_\zeta g_{ab} \quad (r > r_{\max}), \quad (22)$$

where the NP components of $\mathcal{L}_\zeta g_{ab}$ are obtained by applying the formulas of App. A mode by mode to (21) in automated form using the spin-raising and lowering properties of Held's angular derivatives, see Table I.

(WHAT HAPPENED TO THE ZERO MODE)? In Fig. ?? we plot the NP components of our h_{ab} for $r \rightarrow \infty$, and verify numerically the Bondi-like fall-off. In Fig. ??, we plot the rh_{nn} NP-component as $r \rightarrow \infty$, which represents the Bondi-mass aspect [?, App. D].

We also compute h_{uu} at the particle $r = r_0$, where we must now also include the corrector x_{uu} in (22) as in (1). This redshift quantity had already computed by [REF], and we find agreement.

Since we have the full corrector x_{ab} , we of course have the entire metric perturbation in a mode-decomposed form with r -dependent coefficients all the way from the horizon to infinity. We could use this to compute e.g., the self-force on the particle. WHAT IS MISSING HERE IS TO INSERT REFERENCES TO THE RESPECTIVE CODE IN EACH STEP.

V. CIRCULAR ORBIT IN KERR

As our main example, we derive the first order metric perturbation in our asymptotically flat IRG for a circular orbit in Kerr. We could proceed by a mode expansion similar to the Schwarzschild case, where spin-weighted spheroidal functions would now appear. While such an “ (ℓ, m) -mode” scheme is possible, we here describe an “ m -mode” scheme, which seems more promising in general for inclined, elliptical orbits as needed in reality. (However, we will briefly describe some changes necessary in order to generalize the (ℓ, m) -mode scheme to Kerr in Sec. ??, as could be useful for moderately inclined and elliptical orbits).

In the m -mode scheme, quantities are generally expanded as

$${}_sX = \sum_{\omega, m} {}_sX_{\omega m}(r, \theta) e^{-i\omega t + im\varphi}, \quad (23)$$

as opposed to (17). Consequently, the mode amplitudes ${}_sX_{\omega m}(r, \theta)$ in a sense already include an infinite ℓ -summation. The downside is that, of course, ordinary differential equations such as the radial Teukolsky equation now become partial differential equations in (r, θ) . The modes therefore have to be stored as an array of number on a 2-dimensional (r, θ) -grid, and the numerical solution is considerably less straightforward. Fortunately, it turns out that the only “hard” equation in our scheme is the Teukolsky equation.

A. Comoving punctures

The puncture is a local, analytically known approximation to the Detweiler–Whiting singular field near the worldline, and it is only ever needed in a neighbourhood of the particle. We use the high-order construction of Ref. [?]]. It is therefore natural, and in practice essential, to construct it in a frame carried along by the particle rather than in Boyer–Lindquist coordinates. Let γ be the worldline with proper time T and four-velocity u^a , and let $e_A{}^a$ be an orthonormal tetrad along γ whose timelike leg is $e_0{}^a = u^a$ and whose spatial triad is Fermi–Walker transported. Local spatial coordinates X^i are obtained by shooting geodesics orthogonal to γ and then Lie-dragging the resulting labels along the congruence,

$$u^a \nabla_a T = 1, \quad u^a \nabla_a X^i = 0, \quad (24)$$

so that $(\partial_T)^a = u^a$ in a whole neighbourhood of γ and the particle sits at $X^i = 0$ for all T .

The payoff of (24) is that comoving stationarity becomes triviality in coordinates. If a tensor field is stationary in the comoving frame, $\mathcal{L}_u F_{a_1 \dots a_p} = 0$, then its tetrad components obey $\partial_T F_{A_1 \dots A_p} = 0$, and the field is a function of the spatial comoving coordinates alone. The local singular field of a point particle is of exactly this type, and this is why the puncture can be written

in a universal form: its dependence on X^i is fixed once and for all by the local expansion, and every trace of the particular orbit is pushed into the expansion coefficients.

This structure survives for a generic bound orbit. The chart (24) is attached to the worldline, so nothing about it requires the orbit to be circular; what changes is that the worldline now oscillates in r and θ . Writing $q = (q_r, q_\theta)$ for the radial and polar phases, we have $r_p = r_p(q_r)$ and $\theta_p = \theta_p(q_\theta)$, the comoving coordinates become phase-dependent,

$$X = X(x^a; q), \quad Y = Y(x^a; q), \quad Z = Z(x^a; q), \quad (25)$$

still with the particle at $X = Y = Z = 0$, and the puncture coefficients become functions on the orbital two-torus. The local monomial structure is untouched; the only genuinely new step is one further Fourier transform over the torus at the end, which converts the local m -mode into the (m, n, k) frequency modes. A circular equatorial orbit is the degenerate case in which the torus collapses to a point, all coefficients are constant along γ , and a single helical frequency $\omega = m\Omega_\varphi$ survives.

We now specialise to that case, recording the minimum needed to follow the m -mode reduction; the construction is given in full in [PZ: cite the comoving-puncture note], and the singular-field expansion it starts from in Ref. [?].

Write $\psi := \varphi - \Omega_\varphi t$ for the corotating phase and introduce the comoving spatial coordinates

$$X = \frac{r - r_0}{\mathcal{B}}, \quad Y = -r_0 \cos \theta, \quad Z = z_c \sin \frac{\psi}{2}, \quad (26)$$

together with a comoving time T that is a fixed linear combination of t and φ . The constants are fixed by the orbit,

$$\Omega_\varphi = \frac{\sqrt{M}}{a\sqrt{M} + r_0^{3/2}}, \quad \mathcal{B} = \sqrt{\frac{\alpha^2 + \rho_0 - 2}{\rho_0}}, \quad (27)$$

$$\mathcal{D} = \sqrt{\frac{2\alpha + \rho_0 - 3}{\rho_0}}, \quad z_c = \frac{2\sqrt{M}\mathcal{B}}{\sqrt{r_0}\Omega_\varphi\mathcal{D}}, \quad (28)$$

with $\rho_0 := r_0/M$ and $\alpha := a/\sqrt{Mr_0}$. The particle sits at $X = Y = Z = 0$, and the map $(t, r, \theta, \varphi) \mapsto (T, X, Y, Z)$ is invertible in a neighbourhood of the worldline, so tensors may be pulled back to Boyer–Lindquist components by the corresponding Jacobian.

Projected on the Kinnersley tetrad, the puncture then has the universal monomial structure

$$h_{(a)(b)}^P = \sum_{i,j,k,p,q} c_{ijkpq}^{(a)(b)} X^i Y^j Z^k R^p \log^q R, \quad (29)$$

where $R^2 = X^2 + Y^2 + Z^2$ and all of the geometry has been absorbed into the coefficients $c_{ijkpq}^{(a)(b)}$. Two reductions bring (29) into a form whose m -modes can be evaluated in closed form. First, powers of Z beyond the first

are traded for powers of $R^2 - X^2 - Y^2$, which leaves at most one explicit factor of Z . Second, that remaining factor is replaced by the single-valued angular variable $Z' := z_c \sin \psi$, using $Z = \frac{1}{2}Z' + O((Z')^3)$ and truncating at the order of the puncture. The result is

$$h_{(a)(b)}^P = \sum_{i,j,k,p,q} d_{ijkpq}^{(a)(b)} X^i Y^j (Z')^k R^p \log^q R, \quad (30)$$

now with $k, q \in \{0, 1\}$, so that only four types of angular integral can occur.

The m -mode of (30) is an integral over ψ at fixed (X, Y) , and since $R^2 = \varrho^2 + z_c^2 \sin^2(\psi/2)$ with $\varrho^2 := X^2 + Y^2$ is linear in $\cos \psi$, every such integral reduces to a single master function

$$J_{\nu m} = (R\varrho)^\nu \frac{\Gamma(\nu+1)}{\Gamma(\nu+|m|+1)} P_\nu^{|m|}(\chi), \quad \chi := \frac{R^2 + \varrho^2}{2R\varrho}, \quad (31)$$

now with $R^2 = \varrho^2 + z_c^2$, through the dictionary

$$R^p \mapsto J_{p/2, m}, \quad (32a)$$

$$Z' R^p \mapsto \frac{z_c}{2i} (J_{p/2, m-1} - J_{p/2, m+1}), \quad (32b)$$

$$R^p \log R \mapsto \frac{1}{2} \partial_\nu J_{\nu m} \big|_{\nu=p/2}, \quad (32c)$$

$$Z' R^p \log R \mapsto \frac{z_c}{2i} \partial_\nu (J_{\nu, m-1} - J_{\nu, m+1}) \big|_{\nu=p/2}. \quad (32d)$$

The entire m -mode puncture is therefore assembled from associated Legendre functions of a single argument, with no numerical angular quadrature. This is what makes the puncture side of the m -mode scheme cheap enough to be recomputed at every radial grid point.

Returning to (25), the one place where the generic case needs more than a promotion of constants is the master integral. Off the equatorial plane the squared local distance acquires a $\sin \psi$ term, $a(q) - b(q) \cos \psi - c(q) \sin \psi$, but this is brought back to the form $a - \beta \cos(\psi - \delta)$ by a phase shift, so that

$$J_{\nu m}(q) = e^{-im\delta(q)} J_{\nu m}(a(q), \beta(q)), \quad (33)$$

and (31) continues to do all the work. The torus Fourier transform is then applied to the resulting local m -mode.

[PZ: Three bookkeeping points here. (i) The comoving-puncture note is written in mostly-plus signature, whereas this paper is mostly-minus; we need to see where this matters. (ii) I have written $\varrho^2 = X^2 + Y^2$ for the local cylindrical radius because ρ is already the GHP spin coefficient (5). (iii) My puncture note gives (31) without a sign in Sec. V but with a factor $(-1)^{|m|}$ in the summary algorithm of Sec. VID. One of the two is likely wrong; it matters, so it should be checked against the originaly nb before this equation goes in.]

B. Effective stress energy tensor

With the m -modes of the puncture in hand, forming the effective source (3) requires applying the linearized

Einstein operator $\mathcal{E}$ to h_{ab}^P . No new derivation is needed for this: the GHP form of $\mathcal{E}$, and of Teukolsky's source operator $\mathcal{T}^{ab}$ and wave operator $\mathcal{O}$ in (10), are given componentwise in our conventions in [?] and [?, App. G]. All three are built from the background spin coefficients together with the four GHP derivatives $\bar{\mathbf{P}}, \mathbf{P}', \bar{\mathbf{\delta}}, \mathbf{\delta}'$, and are therefore fixed once those derivatives are known.

The m -mode reduction is correspondingly trivial. Every operator entering $\mathcal{E}$, $\mathcal{T}^{ab}$ and $\mathcal{O}$ is diagonal in m and ω , so on a mode of the form $e^{-i\omega t + im\varphi}$ one simply substitutes

$$\partial_t \longrightarrow -i\omega, \quad \partial_\varphi \longrightarrow im \quad (34)$$

wherever they appear, equivalently $\partial_u \rightarrow -i\omega$ and $\partial_{\varphi_*} \rightarrow im$ in the Eddington chart of Table I. The GHP derivatives thereby become first-order differential operators in (r, θ) alone, with m - and ω -dependent coefficients, and the NP components of $\mathcal{E}h_{ab}^P$ follow from the m -modes of Sec. V A by differentiation on the (r, θ) grid. No angular decomposition, and in particular no convolution over m , is involved at any point.

Which components are needed depends on what consumes them. The corrector hierarchy of Sec. V C is sourced entirely by the l -contractions of the effective source, T_{ll}^E , T_{lm}^E together with its conjugate, and T_{ln}^E , which is a reflection of the ingoing radiation gauge. The Teukolsky sources $\mathcal{T}^{ab}T_{ab}^E$ instead require $\{ll, lm, mm\}$ for $s = +2$ and $\{nn, n\bar{m}, \bar{m}\bar{m}\}$ for $s = -2$.

One practical point deserves emphasis, because the obvious shortcut is wrong. The puncture is a *local* construction: the expansion (29) is valid near the worldline, and its extension to the rest of the sphere carries no meaning. It must therefore be localized by the window function *before* $\mathcal{E}$ is applied, so that what is differentiated is already a globally defined field. Windowing after differentiation — multiplying the finished Teukolsky source by a window — gives a different source, and in practice one whose angular behaviour near the poles is badly behaved. We therefore window at the level of h_{ab}^P to produce a regular extension to the poles, and to produce a compact source in the radial direction. [PZ: check refs for $\mathcal{E}$ etc]

C. Integration of GHZ equations for corrector

At the outset, the corrector does not enter the asymptotic argument that is the subject of this paper, however, it is needed to generate a globally (in the exterior region at least) defined “shadowless” metric perturbation. We integrate the GHZ transport equations inward from $\mathcal{S}^+$ with vanishing data there, and the effective source (3) has compact radial support in the worldtube by construction, so x_{ab} vanishes identically from the outer edge of the worldtube out to $r = \infty$. In that whole region the metric is given by (22) with no corrector at all, and the Bondi fall-off established in Sec. III is untouched by anything in this subsection.

The corrector is nonetheless an essential part of the construction, because it is needed to make the metric perturbation shadowless (the generalization for a puncture of no-string for a point source) and must be solved for in the matter region. It factors into the gauge transformation equations of Ref. [?], and it is there that the shadow is removed. This is the puncture-method statement; its counterpart for a point-particle source is the absence of the string-like gauge singularity emanating from the worldline, and the two should not be conflated. The corrector is also needed at the particle itself, where h_{uu} receives the contribution x_{uu} [PZ: check this last point].

In the ingoing radiation gauge the corrector has three independent Newman–Penrose components, and the GHZ equations for them form a triangular system of radial transport equations along the outgoing principal null congruence,

$$\mathcal{T}_1 x_{m\bar{m}} = T_{ll}^E, \quad (35a)$$

$$\mathcal{T}_2 x_{nm} = T_{lm}^E + \mathcal{N}[x_{m\bar{m}}], \quad (35b)$$

$$\mathcal{T}_3 x_{nn} = T_{ln}^E + \mathcal{U}[x_{m\bar{m}}] + \mathcal{V}[x_{nm}], \quad (35c)$$

where $\mathcal{T}_i$ are radial operators and $\mathcal{N}, \mathcal{U}, \mathcal{V}$ are built from the Held operators of Table I; explicit forms are given in [?] and [? , Sec. 3.1.3]. The system is triangular, so the three levels are solved in sequence, each using the derivatives of the fields already obtained.

The m -mode reduction of (35) is again the substitution (34), after which each level is an ordinary differential equation in r at fixed $z = \cos\theta$. The two-dimensional problem therefore decouples into N_z independent radial problems, and this — not the Teukolsky equation — is the sense in which the corrector is the easy part of the m -mode scheme. Two of the three levels are second order in r ; the third is first order once the Held form is expanded,

$$\frac{1}{2}(\rho + \bar{\rho})^2 \partial_r \left[\frac{x_{nn}}{\rho + \bar{\rho}} \right] = \text{source}. \quad (36)$$

Each radial problem is discretized by Chebyshev collocation on two patches meeting at the particle, with interface conditions imposed on the value and the derivative, so that the non-smoothness of the windowed source at $r = r_0$ is never differentiated across.

[PZ: Code references: `solve_worldtube_hierarchy` and `solve_worldtube_level` in `ghz::transport` (`Corrector.hpp`), with the three levels built by `build_xmmbar_two_domain_eq_slice`, `build_xnm_two_domain_eq_slice` and `build_xnn_two_domain_eq_slice` in `TransportEqnBuilders.hpp`; the patched radial solve is `ghz::collocation` (`SpectralSliceSolve.hpp`).]

D. Weyl scalar and Hertz potential

This is the one genuinely hard step of the m -mode scheme, and the only place where a partial differential

equation has to be solved. We record the Teukolsky operator in the Held form we actually use, since it makes both the mode reduction and the structure of the radial problem transparent.

The Teukolsky wave operator $\mathcal{O}$ and source T_0 for ψ_0 take a compact form in Held variables, which we record in Appendix B. In the retarded outgoing chart, in which $\mathbf{P} = \partial_r$ and $\tilde{\mathbf{P}}' = \partial_u - \Delta\rho\bar{\rho}\partial_r$, the resulting equation is

$$\hat{A}\partial_r^2\psi_0 + 2\partial_u\partial_r\psi_0 + \hat{B}\partial_r\psi_0 + \hat{C}\psi_0 = T_0, \quad (37)$$

with coefficients $\hat{A}, \hat{B}, \hat{C}$ given in Appendix B. These are *operator-valued* coefficient functions of (r, z) : only $\hat{A} = -\Delta\rho\bar{\rho}$ is a plain function, while $\hat{B}$ contains $\tilde{\mathbf{D}}$ and $\tilde{\mathbf{D}}'$ separately, and $\hat{C}$ contains them only through the combination $\tilde{\mathbf{D}}\tilde{\mathbf{D}}'$, which is the angular operator that would normally permit separation. Note also that $\hat{C}$ carries no radial derivatives.

The m -mode reduction is again (34). Putting $\partial_u \rightarrow -i\omega$ and $\partial_{\varphi_*} \rightarrow im$ in (37), the mixed derivative and the ∂_u term inside $\hat{C}$ become algebraic, and

$$\hat{A}\partial_r^2\psi_{0,\omega m} + (\hat{B} - 2i\omega)\partial_r\psi_{0,\omega m} + \hat{C}_{\omega m}\psi_{0,\omega m} = T_{0,\omega m}, \quad (38)$$

where $\hat{C}_{\omega m}$ is $\hat{C}$ with ∂_u replaced by $-i\omega$. Because $\tilde{\mathbf{D}}$ and $\tilde{\mathbf{D}}'$ still carry ∂_θ , (38) remains a partial differential equation in (r, θ) at each (m, ω) — this is the “hard” equation referred to at the start of Sec. V, and the only one. We solve it with retarded boundary conditions by a multidomain spectral method [?], with domain boundaries placed at the edges of the worldtube so that the windowed effective source is smooth on each domain. We then verify a posteriori that the solution satisfies peeling (11), so that $\psi^{5\circ}$ may be read off as the coefficient of r^{-5} .

The Hertz potential follows without solving anything further. The inversion relation (12) is $\mathbf{P}^4\Phi = 2\bar{\psi}$ with $\mathbf{P} = \partial_r$ in this chart, so Φ is obtained by integrating $\bar{\psi}$ inward in r four times — in modes, $\bar{\psi}$ at (m, ω) is assembled from ψ at $(-m, -\omega)$ by the conjugation rule quoted with Table II — starting from a radius much larger than both M and the orbital radius and taking vanishing data there. This selects precisely the minimal solution $\Phi = O(r^{-1})$, and the growing pieces that are left undetermined by (12) are exactly the Held integration constants $\Phi^{0\circ}, \dots, \Phi^{3\circ}$ supplied by the completion of Sec. V E.

E. HMS and residual gauge vector

Up to this point the m -mode scheme has had to contend with partial differential equations in (r, θ) . The completion is different: because f° and the four $\Phi^{n\circ}$ are Held scalars, they carry no r -dependence at all, and the entire content of Sec. III at fixed (m, ω) is a problem on the sphere alone. What is lost relative to Schwarzschild is only the ℓ label that made the HMS (14) an algebraic division; what is gained is that nothing has to be resummed afterwards.

Putting $\tilde{\mathbf{P}}' \rightarrow -i\omega$ in (14), the HMS at fixed (m, ω) reads

$$\tilde{\mathbf{D}}'^4 \tilde{\mathbf{D}}^4 \bar{f}_{\omega m}^\circ + 9M^2 \omega^2 \bar{f}_{\omega m}^\circ = 2i\omega^3 \bar{\psi}_{0, \omega m}^{50}, \quad (39)$$

an eighth-order linear ordinary differential equation in $z = \cos \theta$ whose only boundary conditions are regularity at the poles. We solve it by collocation on Legendre–Gauss–Lobatto nodes in z . Two ingredients make this practical.

The first is pole factorization. A spin-weight- s quantity at azimuthal number m carries known endpoint behaviour, and we work throughout with the regular reduced amplitude $\hat{X}$ defined by

$$X = (1 - z)^{|m+s|/2} (1 + z)^{|m-s|/2} \hat{X}. \quad (40)$$

In this representation Held’s angular derivatives act as first-order operators with polynomial coefficients,

$$\tilde{\mathbf{D}} \rightarrow \frac{1}{\sqrt{2}} \left[C_+(m, s, z) \partial_z + B_+(m, s, z) - a\omega C_+(m, s, z) \right], \quad (41)$$

and similarly for $\tilde{\mathbf{D}}'$ with coefficients C_- , B_- and the opposite sign on the undifferentiated term. Here $C_\pm$ and $B_\pm$ are polynomials of degree at most one in z , fixed by the signs of $m \pm s$, and ∂_z is realized by the Legendre differentiation matrix. Because the removed factor in (40) is exactly the one that vanishes at the poles, no quantity in the chain ever has to be evaluated at a coordinate singularity. The GHP weights are carried by the field type rather than supplied by hand, so each application shifts $(p, q) \rightarrow (p, q - 2)$ or $(p - 2, q)$ and the spin weight entering (41) is always the correct one.

The second is that (39) is solved as a dense linear system rather than integrated. The left-hand side is assembled by eight successive applications of (41), four of $\tilde{\mathbf{D}}$ and four of $\tilde{\mathbf{D}}'$; applying this composite operator to each collocation basis vector in turn gives the $N_z \times N_z$ matrix column by column. The system is row- and column-equilibrated, factorized by full-pivot LU, and refined iteratively; the relative residual and the condition number of the equilibrated matrix are returned alongside the solution and are the primary diagnostics of the step.

With $\hat{f}_{\omega m}^\circ$ and its conjugate in hand, the four Held integration constants (15) and the components of ζ^a (16) are formed on the same grid by further applications of (41), with no ℓ projection at any stage. Since Φ^{adj} is a cubic polynomial in ρ^{-1} and $\mathcal{S}^\dagger$ is second order in r -derivatives, each NP component of the adjusted perturbation is returned as a Laurent polynomial

$$h_{ab}^{\text{adj}} = A_{ab}^{(1)} r + A_{ab}^{(0)} + \frac{A_{ab}^{(-1)}}{r} + \frac{A_{ab}^{(-2)}}{r^2}, \quad (42)$$

with the four coefficients stored as arrays over the z nodes, separately for the reconstructed piece, the gauge piece, and their sum. The Bondi fall-off is then simply

the statement that $A^{(1)} = A^{(0)} = 0$ at every collocation node in the nm , nm and mm channels, which can be checked pointwise without any mode sum.

[PZ: Code references for this subsection, all in the `ghz::asymptotic` module of the C++ base `GHZ_numeric`. The reduced operators (41) are `edthHRed_inplace` and `edthBarHRed_inplace` `KinnersleyHeld0Operators`, in `PoleFactorizedSpectralHeld0Operators`. The HMS (39) is `BondiHeldSeedSolver::apply` (the composite operator) and `::solve` (the equilibrated LU with three refinement steps, returning `relative_residual` and `equilibrated_condition_number`). The Laurent coefficients (42) are `BondiHeldMetricReconstruction::reconstruct`, which returns `reconstructed`, `lie_zeta` and `adjusted` for each of `nn`, `nm`, `mm`. The gauge vector and the redshift pullback are `SchwarzschildBondiGauge::summed_reduced_held_mode` and `::lie_derivative_uu`; we should decide how these should be cited — a Zenodo DOI for the repository plus class names in the text is probably cleanest.]

Two independent checks are available at this stage, and both are worth quoting in Sec. V F. The closed-form (ℓ, m) expressions of Sec. IV can be evaluated mode by mode and summed onto the same z grid, giving a reference solution for the seed and for each Laurent coefficient; and the collocation solve of (39) can be compared directly against the exact f° obtained from (19), node by node, which isolates the error of the angular discretization from every other part of the pipeline.

[PZ: Two things to confirm before this subsection is final. (i) The code currently forms `adjusted = reconstructed + lie_zeta`, whereas (22) has a minus sign in front of $\mathcal{L}_\zeta g_{ab}$; the header comment on `SchwarzschildBondiMetric` says this is deliberate and follows the current notebook, but the paper and the code should not differ in sign. (ii) The Bondi driver is at present run at $a = 0$, even though (41) carries the $a\omega$ term and the operators are Kerr-ready. We should either say explicitly that the m -mode completion has so far been validated in Schwarzschild and that Kerr needs only the Kerr ψ_0^{50} data, and run it at $a \neq 0$ before submission.]

F. Numerical results for h_{ab}

We close this section with benchmarks of the m -mode chain. So that every intermediate object can be compared against a closed-form reference, the runs reported here are performed at $a = 0$: the operators, the grid and the reconstruction are the Kerr ones described above, but the background is Schwarzschild, where the (ℓ, m) sums of Sec. IV are available in closed form. All figures use a circular geodesic at $r_0 = 10M$ with $M = 1$, fitted ψ_0 data summed through $\ell = 20$ in the mostly-minus signature, and $N_z = 21$ Legendre–Gauss–Lobatto nodes in $z = \cos \theta$.

Seed inversion. Figure 1 compares the collocation so-

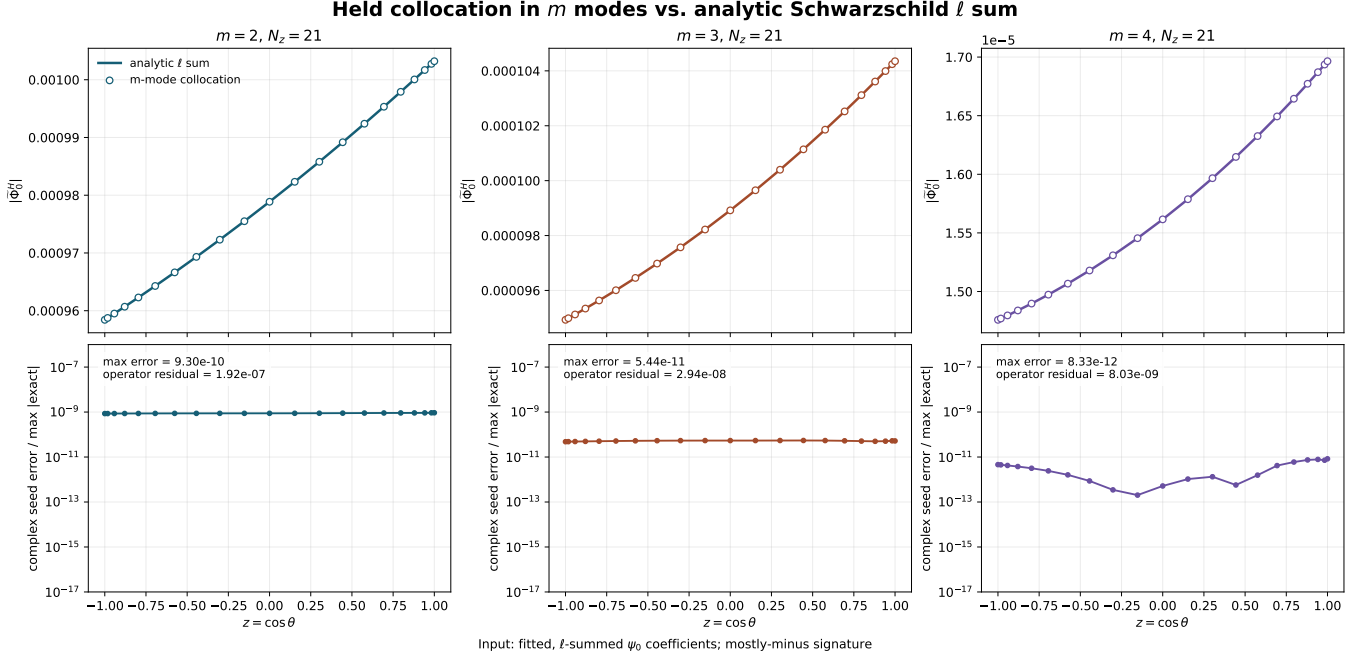


FIG. 1: Validation of the m -mode Held seed. Top row: the modulus of the reduced seed $\tilde{\Phi}^{0o}$ on the $N_z = 21$ LGL grid (circles), obtained by solving (41) in z , against the analytic Schwarzschild ℓ sum (solid). Bottom row: the pointwise complex error relative to $\max |\text{exact}|$. The quoted operator residual is the relative residual of the reduced angular equation when the analytic seed is substituted, and measures the discretization rather than the solve.

lution of the reduced angular equation (41) with the analytic ℓ -summed seed for $m = 2, 3, 4$. The maximum relative complex seed errors are 9.30×10^{-10} , 5.44×10^{-11} and 8.33×10^{-12} , with relative operator-equation residuals for the analytic seed of 1.92×10^{-7} , 2.94×10^{-8} and 8.03×10^{-9} . The errors fall steeply with m , as expected from the pole factorization (40): the residual function $\hat{X}$ becomes smoother as $|m|$ grows, so the spectral representation converges faster. The equilibrated condition number of the discrete operator falls correspondingly, from 8.6×10^7 at $m = 2$ to 4.9×10^5 at $m = 4$, which is why the equilibrated full-pivot solve with iterative refinement is used rather than a plain factorization. This validates both the ℓ -summed input and the m -mode seed inversion.

Cancellation and Bondi falloff. The central claim of Sec. III is that the r^3 and r^2 terms carried separately by $2 \text{Re } S^\dagger \Phi^{\text{Adj}}$ and by $\mathcal{L}_\zeta g_{ab}$ cancel in the sum (22), leaving a perturbation with Bondi-like falloff. Figure 2 exhibits this cancellation in the modal reference. Across 1254 non-stationary radiative mode components through $\ell = 20$, the largest relative cancellation error in the growing and constant Laurent coefficients is 4.90×10^{-16} , i.e. at the level of double precision. Direct finite-radius sums at $z = 0.37$ for $m = 2, 3$ show h_{nn} , h_{nm} and h_{mm} approaching $1/r$ between $r = 20M$ and $r = 2000M$, with fitted large-radius slopes between -1.003 and -1.000 . The residual drift from -1 is the expected contamination from the subleading $\log r/r^2$ and $1/r^2$ terms at finite radius, and shrinks as the fitting window is pushed outward.

The spectral operator chain. Figure 3 repeats the radial test with the full m -mode pipeline in place of the modal reference: the $+m$ and $-m$ seeds are obtained by collocation, the reflected partner being the conjugate of the reduced $-m$ seed since the angular harmonics already supply the odd- m reflection phase, and all subsequent angular derivatives are taken spectrally. At the interior node $z = 0.3020$ the six plotted components again have large-radius slopes between -1.003 and -1.000 . Across all interior nodes the largest relative difference from the modal reference of Fig. 2 is 1.04×10^{-6} for $m = 2$ and 1.54×10^{-7} for $m = 3$, while the largest relative residual in the growing and constant Laurent coefficients is 2.24×10^{-12} . The cancellation is therefore not an artifact of the mode sum: it survives the spectral realization of the Held operators, which is the only form available in Kerr.

Gauge contribution at the particle. The quantity of physical interest at the worldline is h_{uu} , to which the residual gauge vector contributes through $u^a u^b (\mathcal{L}_\zeta g)_{ab}$. For a circular geodesic the modal formula carries an explicit factor of $(m\Omega_\varphi - \omega)$, so this contraction must vanish identically on shell even though $u \cdot \zeta$ itself does not. Figure 4 confirms this: over 70 modes the maximum helical value is 2.42×10^{-19} , at roundoff, while $|u \cdot \zeta|$ ranges over some twelve decades in the same modes. Detuning the frequency by $\delta\omega = 0.004$ raises the maximum to 3.93×10^{-3} , confirming that the null result is the $(m\Omega_\varphi - \omega)$ identity and not an accidental cancellation

Schwarzschild modal adjustment: $2 \operatorname{Re} \mathcal{S}^\dagger(\Phi^{\text{Adj}}) + \mathcal{L}_\zeta g$

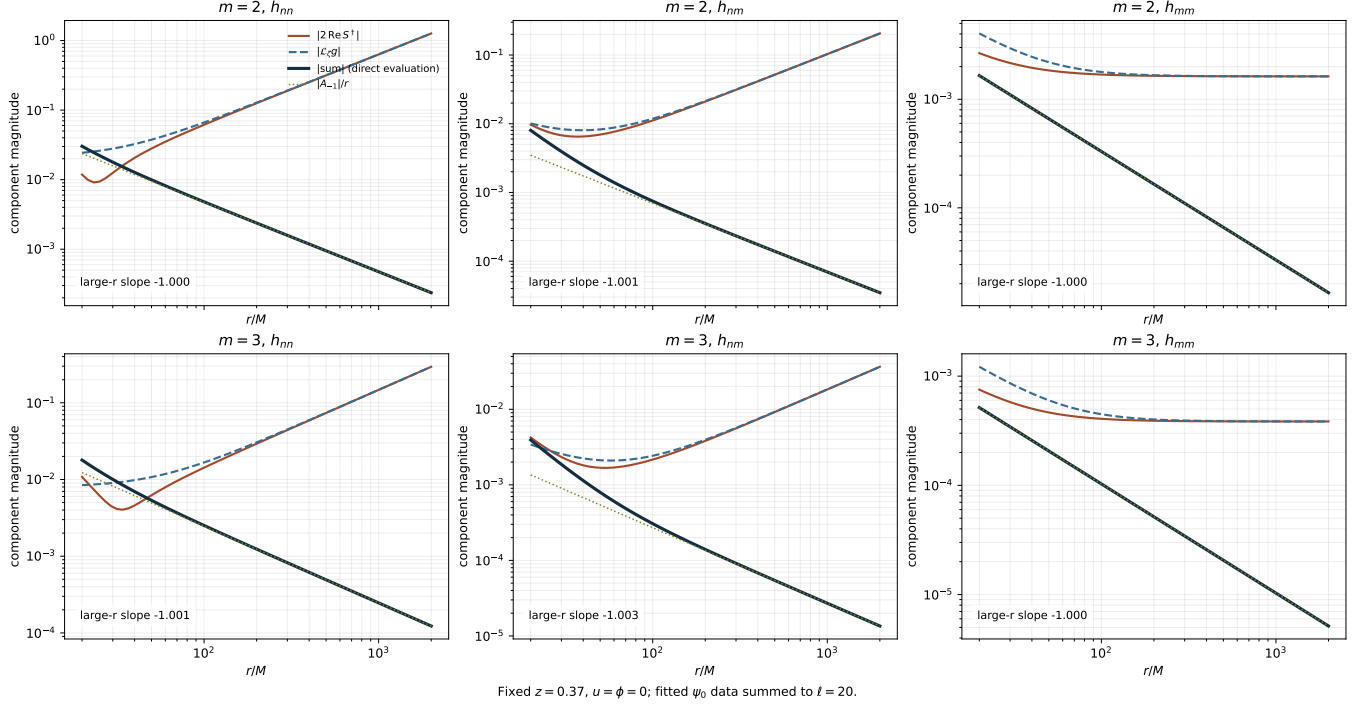


FIG. 2: Independent modal reference for the adjustment (22), built from the closed-form Schwarzschild (ℓ, m) components summed at fixed m through $\ell = 20$, evaluated at $z = 0.37$, $u = \varphi = 0$. Each panel shows the reconstructed piece $|2 \operatorname{Re} \mathcal{S}^\dagger \Phi^{\text{Adj}}|$, the gauge piece $|\mathcal{L}_\zeta g|$, their sum, and the fitted $|A_{-1}|/r$ Laurent coefficient. The individually growing pieces cancel to leave a $1/r$ tail.

or a quantity that was never computed. This checks the gauge-vector contraction only, and is not by itself a validation of the full h_{uu} , which also requires the corrector x_{uu} of Sec. V C.

[PZ: Three things on this subsection. (i) the numbers quoted above are read off the plotting README and `m_mode_summary.csv`. The condition numbers and the m -scaling remark may be deleted. (ii) The original *retarded* Hertz metric is not shown in any of these figures, only the adjusted one. (iii) As flagged above, everything here is $a = 0$.]

G. Remarks on (ℓ, m) -mode scheme in Kerr

For moderately inclined and eccentric orbits an (ℓ, m) -mode treatment remains attractive, since every step of our scheme other than the Teukolsky solve itself stays algebraic. We summarize here what changes relative to the Schwarzschild case of Sec. IV.

A generic bound geodesic has three fundamental frequencies, and any spin- s quantity is expanded as

$${}_s X = \sum_{\ell m n k} {}_s X_{\ell m n k}(r) {}_s S_{\ell, m}(\theta; a \omega_{mnk}) e^{-i \omega_{mnk} u + i m \varphi_*}, \quad (43)$$

in place of (17), where $\omega_{mnk} = m \Omega_\varphi + n \Omega_r + k \Omega_\theta$. The

asymptotic amplitudes $\psi_{0, \ell m n k}^{50}$ are the $\mathcal{S}^+$ amplitudes of the $s = +2$ point-particle Teukolsky solution and require no new radial integrations beyond those already performed. Only the canonical half of the (m, n, k) lattice need be computed, the remainder following from the polar reflection symmetry of the spheroidal functions,

$${}_s Z_{\ell, -m, -n, -k}^\infty = (-1)^{\ell+k} ({}_s Z_{\ell m n k}^\infty)^*. \quad (44)$$

In practice we do not solve the radial Teukolsky equation ourselves in this scheme. The seed data are obtained from the publicly available Julia package `GENERALIZEDSASAKINAKAMURA.JL` [? ?], which integrates the frequency-domain radial Teukolsky equation for Kerr in its generalized Sasaki–Nakamura form. Transforming to the GSN variable removes the long-range part of the Teukolsky potential, so that the asymptotic behaviour is reached at moderate radii and the $\mathcal{S}^+$ and horizon amplitudes are extracted from short-range solutions rather than from slowly convergent series; this is precisely the quantity our scheme consumes, and it is the numerically delicate one at small ω_{mnk} and large ℓ . The package covers $s = 0, \pm 1, \pm 2$ and arbitrary a , uses `SPINWEIGHTEDSPHEROIDALHARMONICS.JL` for the angular problem, and supplies point-particle amplitudes for generic bound orbits parametrized by (p, e, x) , from which the lattice of frequencies ω_{mnk} and the corresponding ${}_s Z_{\ell m n k}^\infty$ follow directly. For each mode of the canonical half-lattice we

Spectral Held m -mode adjustment: $2 \operatorname{Re} S^\dagger (\Phi^{\text{Adj}}) + \mathcal{L}_\zeta g$

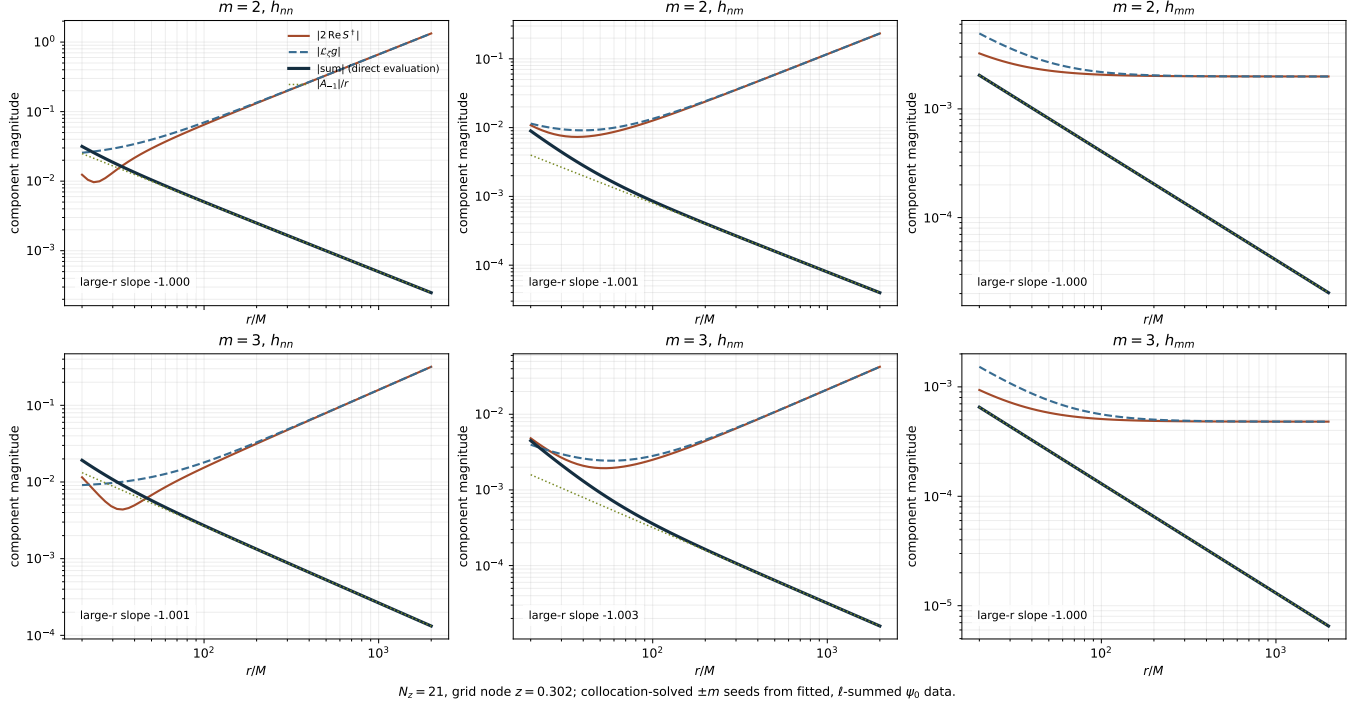


FIG. 3: The same six components as Fig. 2, reconstructed instead from the collocation-solved $\pm m$ Held seeds on the $N_z = 21$ grid and evaluated at the interior node $z = 0.3020$. The reconstruction forms $\Phi^{0\circ}$ through $\Phi^{3\circ}$, applies the spectral Held angular derivatives for $S^\dagger$ and $\mathcal{L}_\zeta g$, and sums the complex Laurent coefficients directly at each radius, without ever projecting back onto individual (ℓ, m) modes.

Schwarzschild circular orbit: gauge contribution to h_{uu}

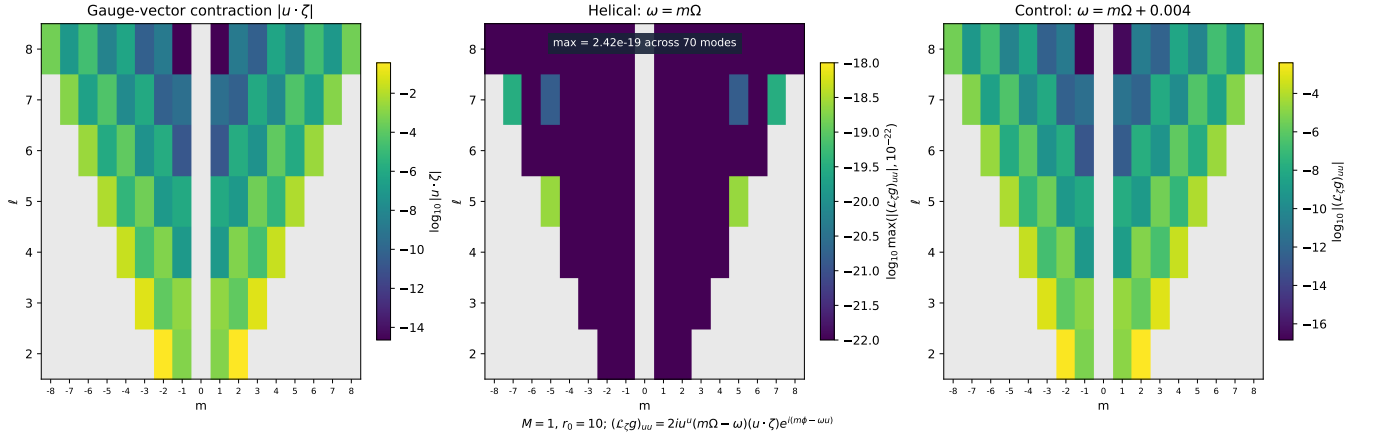


FIG. 4: Gauge contribution to h_{uu} for the Schwarzschild circular orbit, over 70 modes with $2 \leq \ell \leq 8$ and $m \neq 0$.

Left: the worldline contraction $|u \cdot \zeta|$, which is generically nonzero. Center: the helical pullback $|u^a u^b (\mathcal{L}_\zeta g)_{ab}|$ evaluated on shell at $\omega = m\Omega_\varphi$. Right: the same quantity detuned to $\omega = m\Omega_\varphi + 0.004$. Color scales are $\log_{10}$ and differ between panels.

take the $\mathcal{J}^+$ amplitude returned by the package, convert it to our conventions, and identify it with $\psi_{0,\ell mnk}^{5\circ}$; the (ℓ, m) data are then piped into the pipeline of Sec. IV, entering as the seed (19) of the Held hierarchy. Nothing else in the scheme is sensitive to how the radial problem was

solved, so the package may be replaced by any code returning amplitudes in the same normalization. **[PZ: Two things to check: (i) the overall convention/normalization map between the package's $\text{Amplitudes}["I"]$ and our $\psi_{0,\ell mnk}^{5\circ}$ — signature and the ρ^{-4} scaling both enter; (ii)**

the package documentation lists an additional citation for the point-particle/inhomogeneous solutions on top of [?] .]

The HMS is unchanged in form. On a mode $\tilde{\mathbf{P}}' \rightarrow -i\omega_{mnk}$, and the eighth-order angular operator in (14) is precisely the combination appearing in the Teukolsky–Starobinsky identities, so it is diagonalized by the spheroidal harmonics and (19) continues to hold with $D_{\ell m \omega}$ the Kerr Teukolsky–Starobinsky constant. Everything downstream of the seed is likewise unchanged as a set of Held equations. What changes is that these equations are no longer diagonal in ℓ , and there are exactly three sources of mixing.

First, the seed is produced in the spheroidal basis, whereas Held’s angular derivatives act simply on spin-weighted *spherical* harmonics. We therefore expand

$${}_s S_{\ell, m}(\theta; a\omega) = \sum_L b_L^{(s, \ell, m)}(a\omega) {}_s Y_{L, m}(\theta), \quad (45a)$$

$$f_{Lmnk}^\circ = \sum_\ell f_{\ell mnk}^\circ b_L^{(+2, \ell, m)}(a\omega_{mnk}), \quad (45b)$$

and work in the spherical basis thereafter. The mixing coefficients depend only on $(s, \ell, m, a\omega)$ and are computed once per mode and cached, since every Held chain downstream reuses them.

Second, $\tilde{\mathcal{D}}$ and $\tilde{\mathcal{D}}'$ acquire off-diagonal pieces proportional to $a\omega$, and the multiplication operators τ° , $\bar{\tau}^\circ$ and Ω° — which vanish identically when $a = 0$ — are themselves off-diagonal. All five are tridiagonal in ℓ at fixed m .

Third, $\rho^{-1} = -(r - ia \cos \theta)$ is no longer a multiple of the identity. Using

$$\cos \theta {}_s Y_{\ell, m} = c_{\ell+1} {}_s Y_{\ell+1, m} + b_\ell {}_s Y_{\ell, m} + c_\ell {}_s Y_{\ell-1, m}, \quad (46)$$

with

$$c_\ell(s, m) = \sqrt{\frac{(\ell^2 - m^2)(\ell^2 - s^2)}{\ell^2(4\ell^2 - 1)}}, \quad b_\ell(s, m) = -\frac{s m}{\ell(\ell + 1)}, \quad (47)$$

the operator ρ^{-1} is the tridiagonal matrix $\mathbf{R} = -r\mathcal{K} + ia\mathbf{C}$, where $\mathbf{C}$ is the matrix of (46), and the adjusted Hertz potential (13) is evaluated in Horner form,

$$\Phi^{\text{adj}} = \Phi^{0^\circ} + \mathbf{R} \left(\Phi^{1^\circ} + \mathbf{R} (\Phi^{2^\circ} + \mathbf{R} \Phi^{3^\circ}) \right), \quad (48)$$

as a vector over ℓ at fixed (m, n, k) . It is worth noting that the same matrix $\mathbf{C}$ also represents $\Omega^\circ = -2ia \cos \theta$,

so the second and third sources of mixing share their only nontrivial ingredient.

Since each of the five angular operators is tridiagonal, a chain of p of them is banded with half-bandwidth p . The longest chain is the $\tilde{\mathcal{D}}'^3 \tilde{\mathcal{D}}^3$ appearing in Φ^{0° , and (48) adds at most three further neighbours, so Φ^{adj} at output multipole L requires the seed out to $\ell \leq L + 6$, and the seed in turn requires spheroidal amplitudes out to $\ell \leq L + 6 + N_{\text{mix}}$, where N_{mix} is the half-width at which (45a) is truncated. The cost of the completion is then $O(L_{\text{max}})$ per (m, n, k) , which is negligible beside the radial Teukolsky solves that produce its input.

Finally, the two schemes can be compared directly: the (ℓ, m) -mode amplitudes may be resummed onto the same (r, z) grid used in Sec. V E through

$${}_s X_{\omega m}(r, z) = \sum_\ell {}_s X_{\ell mnk}(r) {}_s S_{\ell, m}(\arccos z; a\omega_{mnk}), \quad (49)$$

which provides a mode-by-mode check of the collocation solve against a calculation that shares none of its angular machinery.

[PZ: This subsection is drafted from `BondiGauge_Adjusted_Generic.nb` in `MathematicaNotebooks/EffectiveSource/lm_mode`; (49) is its `ellSummedMode/ellSummedModeRZ`. Open items: (i) quote the production $(\ell_{\text{max}}, n_{\text{max}}, k_{\text{max}})$ and the orbit — the notebook is currently exploratory at $a = 0.667M$, $p = 10M$, $e = 0.1$, $\cos \chi = \cos(\pi/4)$, $\ell_{\text{max}} = 3$, $n_{\text{max}} = 3$, $k_{\text{max}} = 4$; (ii) quote N_{mix} and justify it (the notebook keeps 30 terms with a 10^{-25} chop, which is far more than needed for $a\omega \lesssim 1$ — a plot of $|b_L|$ against $|L - \ell|$ at the largest $a\omega_{mnk}$ in the run would let you state a much smaller effective value); (iii) say whether (44) is meant for general s or only $s = +2$, which is all the notebook uses.]

H. Computational cost

Appendix A: Lie derivative of g_{ab} along ζ^a

Since ζ^a is by construction compatible with the traceless IRG, we have $\mathcal{L}_\zeta g_{ab} l^a = 0$ and $\mathcal{L}_\zeta g_{ab} m^a \bar{m}^b = 0$. The remaining, non-zero, NP components are

$$(\mathcal{L}_\zeta g)_{mm} = -2\bar{\rho} \tilde{\mathcal{D}} \tilde{\mathcal{D}}' \zeta_l^\circ + 2\tilde{\mathcal{D}}' \tilde{\mathcal{D}} f^\circ + 2\Omega^\circ \tilde{\mathbf{P}}' f^\circ + 4\bar{\rho} \tau^\circ \tilde{\mathcal{D}}' f^\circ + 8\rho'^\circ f^\circ, \quad (A1)$$

$$(\mathcal{L}_\zeta g)_{nm} = \bar{\rho} \tilde{\partial}' \tilde{\partial} \tilde{\partial} \zeta_l^\circ - 2\rho \bar{\rho} \bar{\tau}^\circ \tilde{\partial} \tilde{\partial} \zeta_l^\circ + \rho(\rho + \bar{\rho}) \Psi^\circ \tilde{\partial} \zeta_l^\circ - \bar{\rho} \Omega^\circ \tilde{\partial}' \tilde{\partial}' \tilde{\partial} f^\circ + 2\rho \bar{\tau}^\circ \tilde{\partial}' \tilde{\partial} f^\circ \quad (\text{A2})$$

$$+ \frac{1}{\bar{\rho}} \left(-2\bar{\rho} \Omega^{\circ 2} \tilde{\partial}' \tilde{\mathbf{P}}' f^\circ + 2(\rho + \bar{\rho}) \Omega^\circ \bar{\tau}^\circ \tilde{\mathbf{P}}' f^\circ - 2\bar{\rho} \tau^\circ \tilde{\partial}' \tilde{\partial}' f^\circ - (\rho^2 \Omega^\circ + \rho) \Psi^\circ \tilde{\partial}' f^\circ \right) \quad (\text{A3})$$

$$+ \bar{\rho} \bar{\Psi}^\circ \tilde{\partial}' f^\circ - 6\bar{\rho} \Omega^\circ \rho'^\circ \tilde{\partial}' f^\circ + 4\rho \bar{\rho} \tau^\circ \bar{\tau}^\circ \tilde{\partial}' f^\circ + 8\rho \rho'^\circ \bar{\tau}^\circ f^\circ, \quad (\text{A4})$$

and

$$(\mathcal{L}_\zeta g)_{nm} = \text{Re} \left[4\rho \bar{\rho} \bar{\tau}^\circ \tilde{\partial}' \tilde{\partial} \tilde{\partial} \zeta_l^\circ - 4\rho^2 \bar{\rho} \bar{\tau}^{\circ 2} \tilde{\partial} \tilde{\partial} \zeta_l^\circ + 2\rho^2 \Psi^\circ \tilde{\partial}' \tilde{\partial} \zeta_l^\circ + 4\rho^2 \bar{\rho} \Psi^\circ \bar{\tau}^\circ \tilde{\partial} \zeta_l^\circ \right. \quad (\text{A5})$$

$$\left. - 2\tilde{\partial}' \tilde{\partial}' \tilde{\partial}' \tilde{\partial} f^\circ - \frac{2}{\rho} \left(4\Omega^\circ \tilde{\partial}' \tilde{\partial}' \tilde{\mathbf{P}}' f^\circ - 4\rho \bar{\rho} \Omega^\circ \bar{\tau}^\circ \tilde{\partial}' \tilde{\partial}' \tilde{\partial} f^\circ + 4(4 + \bar{\rho} \Omega^\circ - \rho \bar{\rho} \Omega^{\circ 2}) \bar{\tau}^\circ \tilde{\partial}' \tilde{\mathbf{P}}' f^\circ \right) \right. \quad (\text{A6})$$

$$\left. - (8\rho'^\circ + 3\rho \Psi^\circ + 3\bar{\rho} \bar{\Psi}^\circ + 2\rho^2 \Omega^{\circ 2} \Psi^\circ + 8\rho \bar{\rho} \tau^\circ \bar{\tau}^\circ) \tilde{\partial}' \tilde{\partial}' f^\circ + 4\rho^2 \bar{\tau}^{\circ 2} \tilde{\partial}' \tilde{\partial} f^\circ + 4\rho(\rho + 2\bar{\rho}) \Omega^\circ \bar{\tau}^{\circ 2} \tilde{\mathbf{P}}' f^\circ \right. \quad (\text{A7})$$

$$\left. + 8\rho^2 \bar{\rho} \tau^\circ \bar{\tau}^{\circ 2} \tilde{\partial}' f^\circ + 4\rho \bar{\rho} \bar{\Psi}^\circ \bar{\tau}^{\circ 2} \tilde{\partial}' f^\circ - 24\rho \bar{\rho} \Omega^\circ \rho'^\circ \bar{\tau}^\circ \tilde{\partial}' f^\circ + 16\rho^2 \rho'^\circ \bar{\tau}^{\circ 2} f^\circ \right]. \quad (\text{A8})$$

The remaining components follow by conjugation, $\mathcal{L}_\zeta g_{ab} n^a \bar{m}^b = \overline{\mathcal{L}_\zeta g_{ab} n^a m^b}$ and $\mathcal{L}_\zeta g_{ab} \bar{m}^a \bar{m}^b = \overline{\mathcal{L}_\zeta g_{ab} m^a m^b}$, which in mode space are

$$(\mathcal{L}_\zeta g)_{n\bar{m}, \omega\ell m} = -(-1)^m \overline{(\mathcal{L}_\zeta g)_{nm, -\omega\ell -m}}, \quad (\mathcal{L}_\zeta g)_{\bar{m}\bar{m}, \ell m} = (-1)^m \overline{(\mathcal{L}_\zeta g)_{mm, -\omega\ell -m}}. \quad (\text{A9})$$

The relative minus sign between the two is the $(-1)^s$ arising when conjugating a spin-weighted harmonic.

Appendix B: Teukolsky operators in Held form

For ψ_0 the Teukolsky wave operator in (10) reads, in Held form,

$$\begin{aligned} \frac{1}{2}\mathcal{O}\psi_0 = & \frac{1}{2}\rho\bar{\rho}(3\Psi^\circ\rho + 8\rho'^\circ)\psi_0 - \frac{1}{2}\left(\Psi^\circ\rho(3\rho + \bar{\rho}) + 2\bar{\rho}\rho'^\circ + \rho\bar{\rho}(-\bar{\Psi}^\circ + 6\rho\tau^\circ\bar{\tau}^\circ + (\rho'^\circ + \bar{\rho}'^\circ)\Omega^\circ)\right)(\mathbb{P}\psi_0) \\ & + \mathbb{P}\tilde{\mathbb{P}}'\psi_0 + \rho\bar{\rho}\bar{\tau}^\circ(\mathbb{P}\tilde{\mathbb{D}}\psi_0) + \rho\bar{\rho}\tau^\circ(\mathbb{P}\tilde{\mathbb{D}}'\psi_0) - (4\rho + \bar{\rho})(\tilde{\mathbb{P}}'\psi_0) - \rho\bar{\rho}(\tilde{\mathbb{D}}\tilde{\mathbb{D}}'\psi_0), \end{aligned} \quad (\text{B1})$$

and the source is

$$\begin{aligned} T_0 = & 2\bar{\rho}(-2\rho + \bar{\rho})\tau^\circ(\mathbb{P}T_{lm}) - 4\rho(\mathbb{P}T_{mm}) - \mathbb{P}\mathbb{P}T_{mm} \\ & + \bar{\rho}\left(2T_{mm}(2\rho + \bar{\rho}) + 4\bar{\rho}^2\tau^\circ(T_{lm} + T_{ll}(\rho - \bar{\rho})\tau^\circ) + \mathbb{P}\tilde{\mathbb{D}}T_{lm} + 4\bar{\rho}(-\rho + \bar{\rho})\tau^\circ(\tilde{\mathbb{D}}T_{ll})\right. \\ & \left. - 2(2\rho + \bar{\rho})(\tilde{\mathbb{D}}T_{lm}) + \tilde{\mathbb{D}}\mathbb{P}T_{lm} - \bar{\rho}(\tilde{\mathbb{D}}\tilde{\mathbb{D}}T_{ll})\right). \end{aligned} \quad (\text{B2})$$

Both are written with the barred and unbarred Held scalars kept distinct, and we have used $[\mathbb{P}, \tilde{\mathbb{D}}] = 0$ in

(B2).

In the retarded outgoing chart the coefficients of (37) are

$$\hat{A} = -\Delta\rho\bar{\rho}, \quad (\text{B3a})$$

$$\begin{aligned} \hat{B} = & -\left(\Psi^\circ\rho(3\rho + \bar{\rho}) + 2\bar{\rho}\rho'^\circ + \rho\bar{\rho}(-\bar{\Psi}^\circ + 6\rho\tau^\circ\bar{\tau}^\circ + \Omega^\circ(\rho'^\circ + \bar{\rho}'^\circ))\right) \\ & + 2\rho\bar{\rho}(\bar{\tau}^\circ\tilde{\mathbb{D}} + \tau^\circ\tilde{\mathbb{D}}'), \end{aligned} \quad (\text{B3b})$$

$$\hat{C} = \partial_r(\Delta\rho\bar{\rho}) - (4\rho + \bar{\rho})\partial_u + \frac{1}{2}\rho\bar{\rho}(3\Psi^\circ\rho + 8\rho'^\circ) - \rho\bar{\rho}\tilde{\mathbb{D}}\tilde{\mathbb{D}}'. \quad (\text{B3c})$$

The terms containing Δ are the price of the retarded chart: they arise because n^μ has a component along r ,

$n^\mu\partial_\mu r = -\Delta/(2\Sigma)$, and they are the only non-Held-covariant pieces.